MoBI Laboratory In Practice: Examples from 2 studies

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Keywords: MoBI, multimodal data, graphomotor, data acquisition, LSL

Introduction

Multimodal brain/body imaging (MoBI) laboratories offer unique opportunities to capture the complex interplay between neural, physiological, and behavioral processes during data collection (Makeig et al., 2009; Gramann et al., 2011). However, the practical implementation of these systems requires careful consideration of experimental design, data quality control, and the challenges posed by simultaneous multi-device recording. Building upon our previous work establishing a single-computer system that utilizes Lab Streaming Layer (LSL), we present preliminary findings from two ongoing studies that demonstrate the laboratory's capabilities for multimodal data collection. These studies explore two distinct research paradigms: graphomotor control and auditory-motor synchronization. These studies illustrate how synchronized multimodal recordings can provide complementary insights into sensorimotor integration, executive function, and individual differences in motor coordination.

Method

Our MoBI laboratory simultaneously records EEG, eye-tracking, physiological measures (electrocardiography, respiration), and behavioral data using a single-computer LSL framework. All data streams are temporally synchronized at acquisition, eliminating post-collection alignment procedures.

Planned Analysis/Results

Study 1 examines graphomotor control through a battery of writing and symbol substitution tasks. Participants complete digit symbol substitution tests and alphabet writing tasks while we record neural activity (32-channel EEG), eye movements (wearable eyetracking), and physiological responses. This paradigm allows investigation of the neural correlates of fine motor control, visual-motor integration, and cognitive processing speed (Decker et al., 2011; Alamargot et al., 2007). The synchronized multimodal data enables us to examine how attention allocation (via eye-tracking), neural oscillations (via EEG), and physiological arousal relate to graphomotor performance and execution speed.

Study 2 implements a speech-to-speech synchronization protocol designed to classify participants as high or low auditory-motor synchronizers. Participants listen to rhythmic speech patterns and attempt to synchronize their vocal responses. We record EEG to capture auditory processing and motor preparation, audio streams to quantify synchronization accuracy, physiological measures to assess arousal and effort, and eye-tracking to monitor attention. This protocol builds upon established auditory-motor coupling paradigms (Repp & Su, 2013; Dalla

Bella et al., 2017) while leveraging our multimodal approach to identify neural and physiological signatures that distinguish synchronization ability.

Preliminary data from both studies demonstrate successful integration of multiple data modalities with precise temporal alignment. Initial analyses reveal interpretable patterns in the multimodal data streams, including task-related changes in EEG spectral power, systematic eye movement patterns during graphomotor tasks, and individual differences in autonomic responses during synchronization attempts.

Our quality control procedures, enabled by the consolidated LSL data format, identified several technical issues during pilot testing that would have resulted in unusable datasets had they gone undetected until analysis. The single-file data structure facilitated rapid preprocessing pipelines and allowed for efficient iteration on experimental protocols.

Conclusion

These two studies demonstrate the practical utility of our centralized MoBI infrastructure for investigating sensorimotor control and coordination. The synchronized multimodal recordings provide rich, complementary data that would be difficult to obtain using traditional single-modality approaches. Our experience highlights both the advantages of LSL-based data collection and important practical considerations for researchers implementing similar systems. Future work will present complete datasets and multimodal analysis pipelines for both paradigms.

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